Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 04-Apr-2026

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1. Administrative & Study Identification

Full Study Title: An Open-label Study to Assess Safety and Efficacy of SZC in Paediatric Patients With Hyperkalaemia **Short Title/Acronym:** PEDZ-K **Protocol Number:** D9481C00001 **NCT Number:** NCT03813407 **Posting ID:** 225014692632882216 **Sponsor Name:** AstraZeneca **Trial Status:** Recruiting **Investigational Product:** Sodium Zirconium Cyclosilicate (SZC) **Trade Name:** Lokelma **ATC Code:** V03AE10 **EMA URL:** Lokelma EPAR Product Information **Phase of Development:** Phase 3 **Medical Monitor:** AstraZeneca Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the ability to achieve normokalaemia during the Correction Phase (CP) within 3 days when initiating treatment with SZC of different dose levels in children with hyperkalaemia, and to evaluate the ability to maintain normokalaemia during the 28-day Maintenance Phase (MP). **Secondary Objectives:** To evaluate the change in serum potassium ($sK+$) in children treated with SZC across all study phases, as well as the safety and tolerability of the drug in a paediatric population.

2.2 Background & Rationale Sodium zirconium cyclosilicate (SZC) has been shown to be effective and safe in adults for the treatment of hyperkalaemia. This study is conducted to address the medical necessity of evaluating the efficacy, safety, and tolerability of SZC utilizing body-weight equivalent dosing regimens to expand the benefit of this potassium binder to the paediatric population (children < 18 years of age) suffering from hyperkalaemia.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A (Sequential age cohort and dose-level enrolment recommended by an independent Data Monitoring Committee)

3.2 Planned Enrollment & Duration Target N: Approximately 140 participants **Number of Sites:** Approximately 46 sites across 11 countries (including Europe and North America) **Estimated Study Start:** Q4 2018 **Estimated Primary Completion:** 27-Dec-2024

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Female or male from birth to < 18 years of age. 2. Confirmed diagnosis of hyperkalaemia requiring short- or long-term treatment. Hyperkalaemia is defined by laboratory thresholds based on age (e.g., $sK+ > 5.0$ mmol/L for age ≥ 2 years; $sK+ > 6.0$ mmol/L for 1 month to < 2 years; $sK+ > 6.5$ mmol/L for 0 to < 1 month). 3. Ability to have repeated blood draws or effective venous catheterisation.

4.2 Exclusion Criteria 1. Neonates with a gestational age < 37 weeks at birth or a birth weight < 2500 g. 2. Term and preterm neonates with suspected conditions predisposing them to intestinal ischaemia (e.g., perinatal hypoxia or sepsis). 3. Participants with pseudohyperkalaemia caused by excessive fist clenching to enable venepuncture, haemolysed blood specimens, or severe leukocytosis/thrombocytosis.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * *Correction Phase (CP):* Fixed dose of SZC three times daily (TID) for up to 3 days (dose scaled to body weight equivalent to an adult 5 g, 10 g, or 15 g). * *Maintenance Phase (MP):* Once daily (QD) administration for 28 days, titrated between 2.5 g to 15 g body weight equivalent to maintain normokalaemia. * *Long-term Maintenance Phase (LTMP):* Same titration regimen as MP.

Route of Administration: Oral **Duration of Treatment:** Approximately 28 weeks (Up to 3 days CP + 28 days MP + up to 22 weeks LTMP).

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care for underlying conditions causing hyperkalaemia (e.g., CKD, heart failure) as long as it aligns with study protocol. **Prohibited:** Therapies known to cause transient iatrogenic hyperkalaemia or non-absorbable antibiotics for hyperammonemia (e.g., lactulose, rifaximin) within 7 days prior to treatment.

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6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening** | Day -14 to 0 | Informed Consent/ Assent, Medical History, $sK+$ Labs Confirmation | | **Correction Phase (CP)** | Days 1 to 3 | First Dose (TID), frequent $sK+$ monitoring until normokalaemia is achieved | | **Maintenance Phase (MP)** | Days 4 to 31 | Once daily (QD) SZC dosing, Safety Labs, Dose titration to maintain normokalaemia | | **LTMP** | Up to 22 Weeks | Continued dose titration, ongoing safety & efficacy monitoring | | **End of Study** | 1 Week Post-Dose | Safety follow-up visit, Exit Interview, IP Accountability |

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: 1. *Correction Phase:* Normokalaemia achieved in the CP within 3 days (yes/no). 2. *Maintenance Phase:* Serum potassium ($sK+$) value within normokalaemia range (yes/no) at each of the last two scheduled visits in the 28-day MP. **Measurement Method:** Central/Local Laboratory Analysis of Serum Potassium.

7.2 Secondary & Exploratory Endpoints * Change in $sK+$ levels in children treated with SZC across all study phases. * Safety and tolerability endpoints including Adverse Events (AEs) and Serious Adverse Events (SAEs).

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via onsite visits and laboratory analysis monitoring across CP, MP, and LTMP. **Serious Adverse Events (SAEs):** Standard reporting timeline (typically 24 hours). **Data Monitoring Committee (DMC):** An independent Data Monitoring Committee (iDMC) reviews all available safety and efficacy data to recommend the opening of higher dose levels and the sequential opening of younger age cohorts (2 to < 6 years and 0 to < 2 years).

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Defined for each phase as the set of all participants who transitioned from the previous phase and received at least one dose of SZC during the specific phase. **Sample Size Justification:** Approximately 140 participants enrolled to adequately evaluate pharmacodynamics, safety, and dosing equivalents in pediatric subsets. **Handling of Missing Data:** Primary assessments of the probability to achieve and maintain normokalaemia will be based on point estimates and 95% confidence intervals (CIs) from generalised linear models (utilizing a repeated measures model for the MP).

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure protocol compliance, especially considering pediatric vulnerability. **Audit Trail:** Strict documentation of dose adjustments (titration), laboratory values, eCRF locking, and data clarification forms.

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11. Study Identifiers & Governance

Primary ID: D9481C00001 **Registry Number:** NCT03813407 **Posting ID:** 225014692632882216 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** PEDZ-K **Official Regulatory Title:** An Open-label Study to Assess Safety and Efficacy of SZC in Paediatric Patients With Hyperkalaemia

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12. Clinical Framework (Hyperkalemia Specific)

Primary Condition: Hyperkalaemia in paediatric patients **Disease Definition:** Abnormally high potassium concentration in the blood, most often due to defective renal excretion. It is characterized clinically by electrocardiographic abnormalities (elevated T waves and depressed P waves, and eventually by atrial asystole). In severe cases, weakness and flaccid paralysis may occur. (Dorland, 27th ed) **Diagnosis Threshold:** $sK+ > 5.0$ mmol/L (for ages ≥ 2 years); age-adjusted for infants **Disease Classifications:** MeSH: D006947 | SNOMED ID: 14140009 | HPO: HP: 0002153 **Medical Methodology:** Potassium binder intervention (SZC / Sodium Zirconium Cyclosilicate) **Optimization Target:** Achievement and maintenance of normokalaemia across age-adjusted body weight equivalent doses

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13. Study Procedures & Methodology

Management Pathway: Three-phase treatment (Correction, 28-day Maintenance, and Long-Term Maintenance) **Education Protocol (HCP):** Site initiation and iDMC-guided dose escalation training for pediatric administration. **Education Protocol (Patient):** Informed assent and caregiver administration training for oral suspension. **Quality Audit Mechanism:** iDMC oversight for age cohort unblinding and dose escalation safety.

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14. Observation & Follow-Up Schedule

Total Duration: Approximately 28 weeks **Onsite Visit Cadence:** Intensive monitoring during 3-day CP, routine scheduled visits during 28-day MP, and spaced visits during the 22-week LTMP. **Remote Monitoring Frequency:** As per specific site protocols for maintenance. **Checklist Review Interval:** Regular interval reviews culminating in a final safety follow-up 1 week after the last dose.

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15. Sign-off & Approval

| Role | Name | Signature | Date | | :--- | :--- | :--- | :--- | | **Principal Investigator** | [Name] | ____ **| [DD-MMM-YYYY] | |**

Clinical Operations Lead | [Name] | ___ | ***[DD-MMM-YYYY]* | |** ***Lead Statistician* | *[Name]* |** _____ | [DD-MMM-YYYY] |